

ATAM

LG Architecture Training Program

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Architecture evaluation

- The goal is to assess to what degree an architecture is fit for the purpose for which it is intended
- The output of the evaluation is identification of risks
- The cost of evaluation needs to be less than the value it provides

If you think good architecture is expensive, try bad architecture.



Brian Foote

How does this architecture address my quality attribute requirements?



The ATAM*

- The Architecture Tradeoff Analysis Method (ATAM) was developed by the Software Engineering Institute (SEI)
- We can use ATAM to assess the consequences of architectural decisions in light of quality attribute requirements and business goals

* This and following slides based on ATAM presentation created by the SEI and used in the ATAM itself

Purpose of the ATAM ₍₁₎

- ATAM is a method that helps stakeholders ask the right questions to discover potentially problematic architectural decisions
- Discovered risks can then be made the focus of mitigation activities—for example, further design, further analysis, and prototyping
- Tradeoffs can be explicitly identified and documented

Purpose of the ATAM ₍₂₎

- The purpose of the ATAM is **NOT** to provide precise analyses
- The purpose **IS** to discover risks created by architectural decisions
- We want to find trends: correlations between architectural decisions and predictions of system properties

ATAM is a scenario-based architecture evaluation method that provides qualitative assessments of risks and tradeoffs

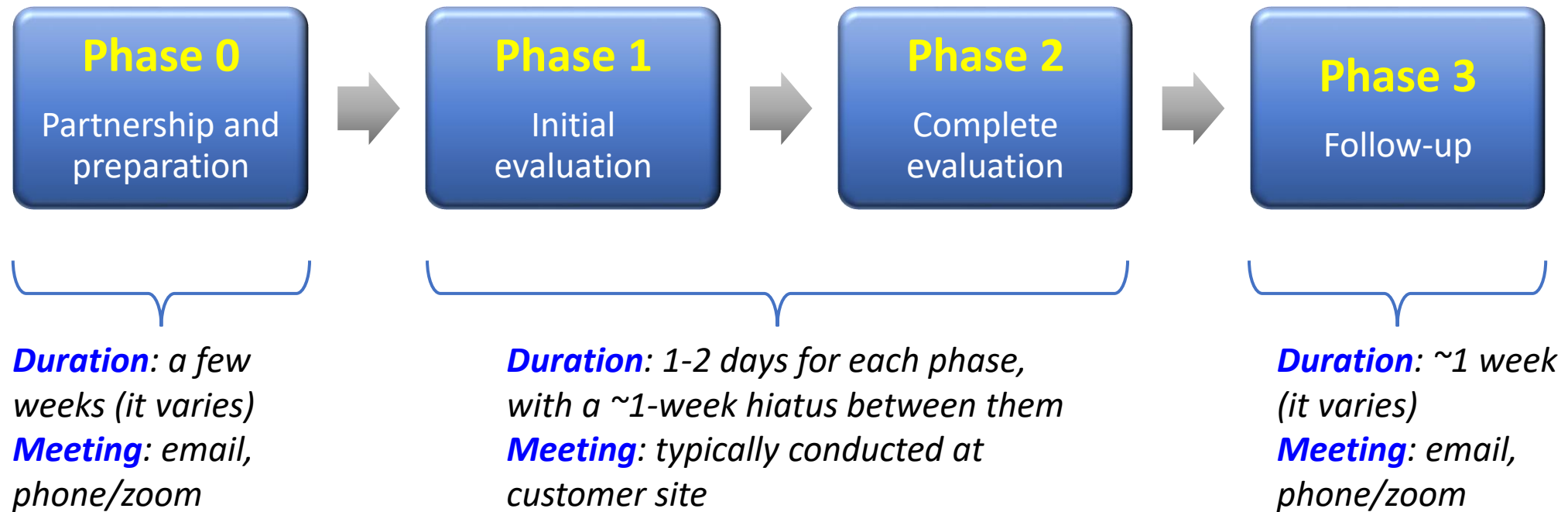
ATAM Benefits

- Identified risks
- Clarified quality attribute requirements
- Improved architecture documentation
- Documented basis for architectural decisions
- Increased communication among stakeholders

At the end ATAM helps to find a successful architecture

ATAM phases

- ATAM evaluations are conducted in 4 phases:



ATAM Phase 0

- This phase precedes the technical evaluation
- During it, the customer and a subset of the evaluation team
 - exchange their understanding about the method and the system whose architecture is to be evaluated
 - come to an agreement about performing the evaluation
 - field a core evaluation team



ATAM Phase 1

- This phase involves a small group of predominantly technically-oriented stakeholders
- Phase 1 is
 - architecture-centric
 - focused on eliciting high priority quality requirements
 - focused on analyzing the architecture fitness to address these requirements

The 9 steps of the ATAM

1. Present the ATAM
2. Present the business goals
3. Present architecture
4. Identify architectural approaches
5. Generate quality attribute utility tree
6. Analyze architectural approaches

Phase 1

7. Brainstorm and prioritize scenarios
8. Analyze architectural approaches
9. Present results

Phase 2

Step 1: Present the ATAM

- The evaluation team presents an overview of the ATAM including
 - the ATAM steps in brief
 - techniques
 - utility tree generation
 - architecture elicitation and analysis
 - scenario brainstorming/mapping
 - outputs
 - architectural approaches
 - utility tree and scenarios
 - risks, non-risks, sensitivity points, and tradeoffs



Step 2: Present the business goals

- The ATAM customer representative describes the system's business goals including the
 - business context for the system
 - high-level functional requirements
 - high-level quality attribute requirements
 - architectural drivers: quality attributes that “shape” the architecture
 - critical requirements: quality attributes that are most central to the system's success



Step 3: Present architecture

- The architect presents an overview of the architecture including
 - technical constraints such as runtime environment (e.g., AWS cloud), hardware, or middleware prescribed for use
 - other systems with which the system must interact
 - architectural approaches used to address quality attribute requirements
- The evaluation team begins probing for and capturing risks



lock | #273469972

Step 4: Identify architectural approaches

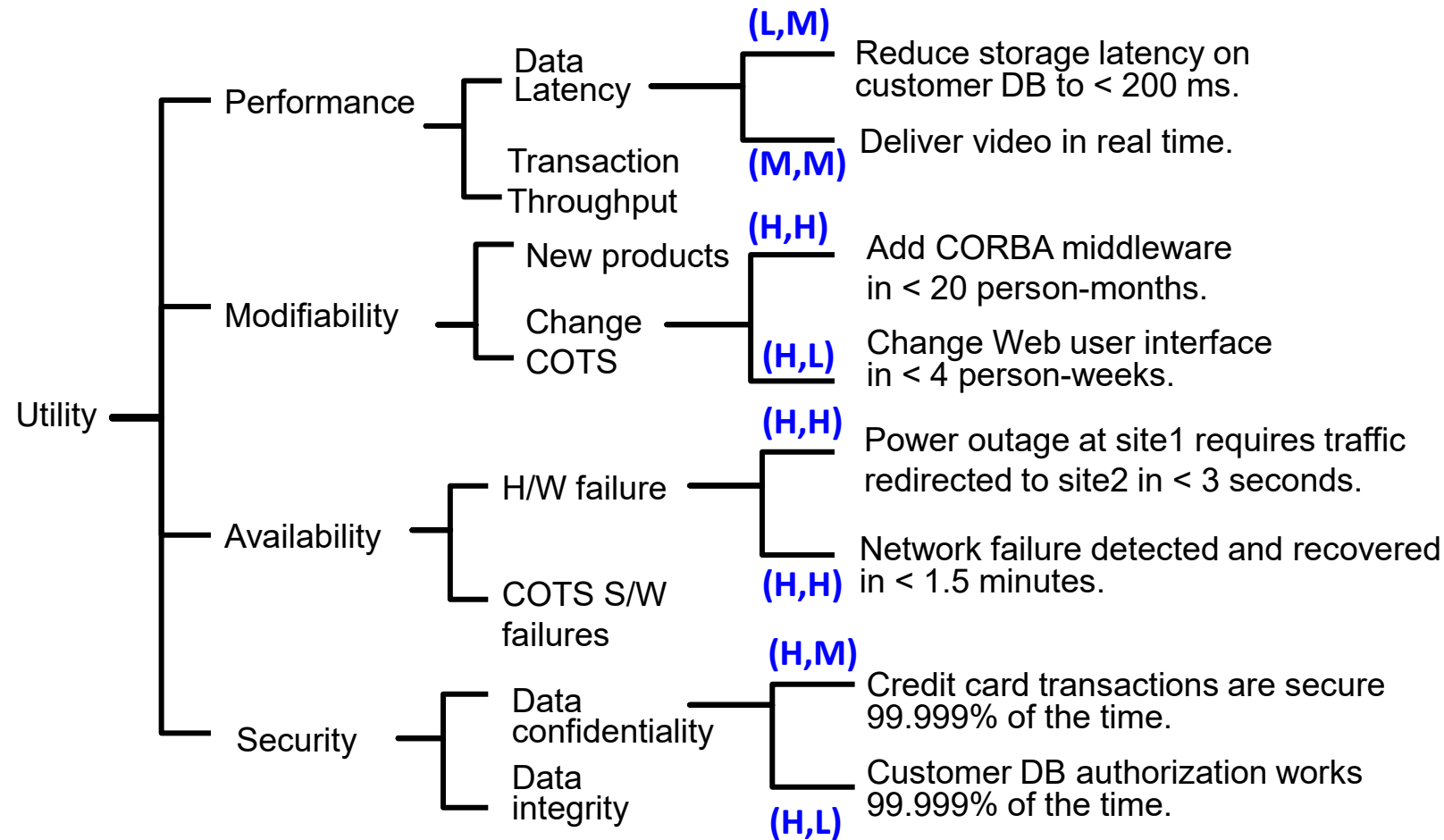
- Identify predominant architectural approaches
- An architectural approach can be the use of a tactic, design pattern, architecture style, or any design strategy. Examples:
 - Ping-echo
 - Passive redundancy
 - Event-driven communication
 - Data replication with ETL synchronization
- Evaluators begin to identify places in the architecture that are key to realizing the system's quality attribute goals

Step 5: Generate quality attribute utility tree

- Identify, prioritize, and refine the most important quality attribute goals by building a **utility tree**
 - A utility tree is a top-down vehicle for describing and prioritizing QA requirements
 - The quality attributes are the high-level nodes (performance, modifiability, security, etc.)
 - Quality attribute scenarios are the leaves of the utility tree
- Prioritization uses two dimensions:
 - Technical difficulty or risk
 - Business importance

The evaluation team can seed the discussion with general scenarios

Utility tree



In practice the utility tree is recorded in a text document in tabular form

Scenarios are checked to make sure they have stimulus, environment, and testable or measurable response

Scenarios rated with respect to (technical difficulty or risk, business importance)

Step 6: Analyze architectural approaches

- The evaluation team probes architectural approaches from the point of view of specific quality attributes to identify risks
- The team
 - identifies the architectural approaches
 - asks quality-attribute-specific questions for the highest priority scenarios
 - identifies and records risks, non-risks, sensitivity points, and tradeoffs

Quality attribute questions

- Quality attribute questions probe architectural decisions that bear on quality attribute requirements
- Performance
 - How many instances of service X should exist at peak load?
 - What are the message-arrival rates?
- Modifiability
 - Are there places where layers are circumvented?
 - Can we change the API of module Y if necessary?

Risk, non-risk, sensitivity point, tradeoff point

- A **risk** is a potentially problematic architectural decision
- A **non-risk** is a good architectural decision that is frequently implicit in the architecture
- A **sensitivity point** is a property of one or more components (and/or component relationships) that is critical for achieving a particular quality attribute response
- A **tradeoff point** is a property that affects 2+ quality attributes, but one of them improves while the other degrades

Examples

- Example risk:
 - *Rules for defining the functional scope of business logic in modules of the domain layer are not clear; this issue may result in replication of functionality, thereby compromising modifiability*
- Example tradeoff:
 - *Using a cloud-based IdP service to authenticate users improves security and modifiability, but will create a performance overhead for on-premises applications that will interact with the service*
- Example sensitivity point:
 - *The number of person-days of effort to create the new REST service is sensitive to the choice between to use Python and the Django framework, which are new to the team, or Java and Spring, which the team is well familiar with*
- Example of non-risk:
 - *Assuming message arrival rates of once per second, a processing time of less than 30 ms, and the existence of one higher priority process, a 1-second soft deadline seems reasonable*

ATAM Phase 2

- This phase involves a larger group of stakeholders
- Phase 2 is
 - stakeholder-centric
 - focused on eliciting the points of view of diverse stakeholders and verifying the Phase 1 results

Stakeholders in phase 2

- In phase 2 we invite a wide group of **stakeholders** (anyone concerned with aspects of the system affected by its architecture)
 - Project manager
 - Developer
 - Product owner/business analyst/domain expert
 - Architect/technical leader
 - Tester/integrator
 - UI designer/Web designer/UX expert
 - Information security office team member
 - Data administrator/DBA
 - SRE/network admin/system admin
 - Device hardware expert
 - Etc.



The 9 steps of the ATAM

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Phase 2

Phase 2 begins with a phase 1 recap

- There are many new participants (broader stakeholder constituency)
- Step 1 is repeated
- Evaluation leader recaps the results of steps 2-6
- At that point everybody is up to speed with the current list of risks, architectural approaches, and tradeoffs



Step 7: Brainstorm and prioritize scenarios

- Stakeholders generate scenarios using a facilitated **brainstorming** process
 - Scenarios at the leaves of the utility tree serve as examples to facilitate the step
- After brainstorming, each stakeholder is granted a number of votes equal to 30% of number of scenarios, rounded up
- Stakeholders can allocate their votes to scenarios in any way they see fit
- The voting-based prioritization can be compared with the utility tree

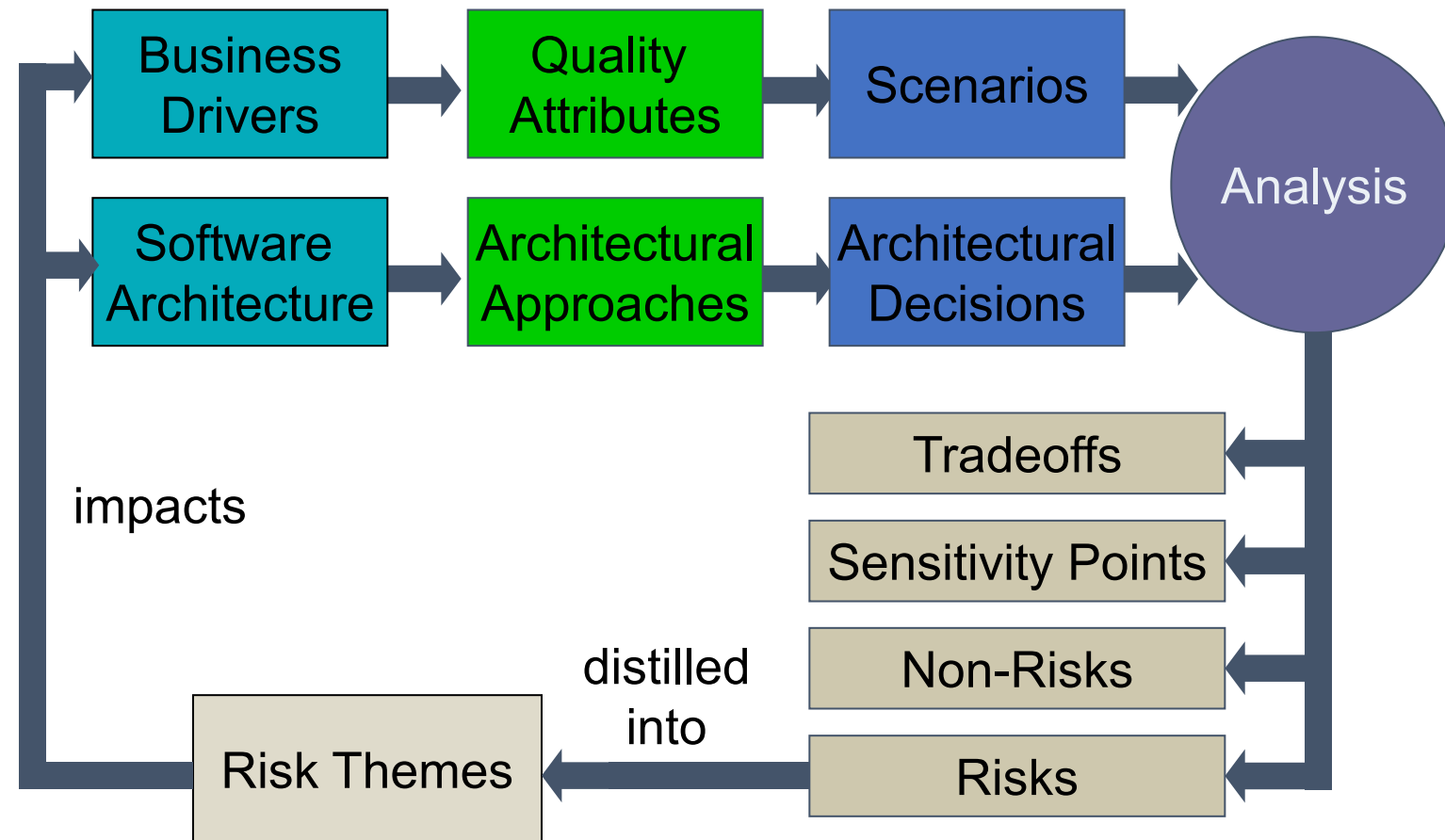
Step 8: Analyze architectural approaches

- Identify the architectural approaches impacted by the scenarios generated in the previous step
 - This step continues the analysis started in Step 6 using the new scenarios
 - Risks and non-risks continue to be identified
 - Architectural information continues to be annotated
- Only top priority scenarios are analyzed, typically 5-10 scenarios

Step 9: Present the results

- Recapitulate all the steps of the ATAM and present the ATAM outputs, including
 - architectural approaches
 - utility tree
 - scenarios
 - risks and non-risks
 - sensitivity points and tradeoffs
 - risk themes

Conceptual flow of the ATAM



ATAM Phase 3

- This phase primarily involves producing a final report for the customer, as well as reflecting on the quality of the evaluation and the ATAM materials
 - If no final report was called for in the ATAM evaluation agreement, report writing is omitted
 - But at least a slide presentation is typically created



The final report/presentation

- The evaluation team will typically create a final report/presentation that includes
 - an executive summary
 - a description of the ATAM
 - a description of the business drivers and architecture
 - the Phase 1 and Phase 2 scenarios and the utility tree
 - the Phase 1 and Phase 2 analysis: architectural approaches, decisions, risks, sensitivities, tradeoffs, and non-risks
 - the risk themes
 - the next steps

*GenAI is limited
for architecture
evaluation...*

... because [Ivers25]:

- *GenAI is limited for reasoning about collections of decisions*
- *GenAI is poorly suited to drawing context-specific conclusions*
- *Evaluation relies on contextualization and high-fidelity representation, which are challenging for GenAI*
- *ASRs often lack information to allow correlation with architecture representations*





Questions?

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